

B. Tech
(SEM V) THEORY EXAMINATION 2022-23
FERMENTATION BIOTECHNOLOGY

Time: 3 Hours**Total Marks: 100**

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief. 2 x 10 = 20

- (a) Differentiate between upstream and downstream processing.
- (b) How microbial activity is optimized?
- (c) What are the criteria for transfer of inoculum?
- (d) What are the advantages of fed batch culture?
- (e) What are the main components of culture media?
- (f) Why antifoams are used in fermenter?
- (g) How Pasteur effect is different from Crabtree effect?
- (h) What do you understand by diauxic growth curve?
- (i) What are the advantages of microbial ore leaching?
- (j) Why microbial cells are ideal choice for biotransformation?

SECTION B

2. Attempt any three of the following: 10 x 3 = 30

- (a) What are the selection criteria for any microbial strain in fermentation industry? What are the problems often associated with industrial microbial processes? How these problems are tackled?
- (b) Explain the material balance for batch and continuous cultures.
- (c) What are the different methods of air sterilization. Describe.
- (d) What are the characteristics of secondary metabolites? Explain the over-production of secondary metabolites with its regulations.
- (e) How microbes are used to produce enzymes? Explain the selection process of fermentation type used for industrial enzyme production with suitable example.

SECTION C

3. Attempt any one part of the following: 10 x 1 = 10

- (a) Why strain improvement is a key to successful fermentation process? How the improved strains are described as GRAS (genetically regarded as safe)? Enlist the pros and cons of different strain improvement methods.
- (b) Describe the evolution of fermentation industry. Briefly explain the effect of introducing modern tools (such as genome editing and CRISPR technology) on the current fermentation industries.

4. Attempt any *one* part of the following: 10 x 1 = 10

- (a) What is biphasic growth? Describe the factors which influence the microbial growth curve.
- (b) Describe-
 - (i) Aseptic inoculation and sampling
 - (ii) Microbial growth measurement methods

5. Attempt any *one* part of the following: 10 x 1 = 10

- (a) Explain the direct and indirect methods of media sterilization.
- (b) Describe-
 - (i) thermal death rate of microorganisms
 - (ii) oxygen requirements of fermentations

6. Attempt any *one* part of the following: 10 x 1 = 10

- (a) What do you understand by operons? Explain the inducible and repressible operons with suitable example.
- (b) Differentiate between feedback inhibition and feedback repression with examples.

7. Attempt any *one* part of the following: 10 x 1 = 10

- (a) What are the types of insulin? How human insulin genes are recombined with microbe? What is proinsulin method? What are the optimal parameters for human insulin production?
- (b) Explain the conventional method of citric acid production. How it is different from integrated citric acid-methane fermentation process?